04-01-2025

① Does a TTC always exist?

Proof:

$\exists\ \mathcal{A} = \{1, \ldots, N\}$, $\mathcal{H} = \{h_1, \ldots, h_N\}$

$a_1 = n-1$ options, $a_2 = n-2, \ldots, a_N = 0$

since they have run out of choices

$\therefore$ a cycle must exist. (Pigeon-Hole)

② Can an agent take part in multiple?

Proof:

If part of 2 cycles, can only go 1 place bc prefs are strict, if 2 cycles point, you will break down once agent i is matched

TTC:

↳ Always runs

↳ Always terminates

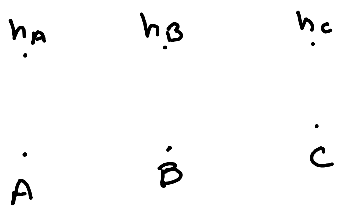
↳ IR

↳ Pareto

$\Big\}$ conditional on reporting truthfully

TTC - Core:

↳ Any allocation w/ no group of agents which would decline allocation to trade amongst selves w/ initial homes



$A: h_B \succ h_C \succ h_A$

$B: h_A \succ h_C \succ h_B$

$C: h_A \succ h_B \succ h_C$

$A \to h_C$; $B \to h_A$; $C \to h_B$ $\Rightarrow$ not in core

↳ IR, Pareto Efficient but A+B exit and privately trade $\therefore$ not in core $\Rightarrow$ blocking coallition

↳ core is set of allocations for which $\nexists$ a blocking coallition

↳ core: cannot be blocked by any subset

↳ TTC finds unique allocation in core